

Experiment 16

Question: Experiment to understand EDA - Quantitative and Qualitative analysis.

Aim

To perform Exploratory Data Analysis (EDA) using quantitative (numerical) and qualitative (categorical) analysis.

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

def exp_16():
    rng = np.random.default_rng(6)
    n = 200
    df = pd.DataFrame({
        "Age": rng.integers(18, 60, n),
        "Salary": rng.normal(50000, 12000, n).round(0),
        "Experience": rng.integers(0, 30, n),
        "Dept": rng.choice(["IT", "HR", "Sales", "Finance"], n, p=[0.4, 0.2, 0.25, 0.15]),
        "Gender": rng.choice(["Male", "Female"], n),
        "Satisfaction": rng.choice(["Low", "Medium", "High"], n, p=[0.2, 0.5, 0.3]),
    })
    df["Salary"] += df["Experience"] * 800      # make Salary depend on Experience

    quant = ["Age", "Salary", "Experience"]
    qual = ["Dept", "Gender", "Satisfaction"]

    print("==== QUANTITATIVE ANALYSIS (numerical data) =====")
    print(df[quant].describe().round(2))
    print("\nMedian :", df[quant].median().to_dict())
    print("Mode of Age :", df["Age"].mode()[0])
    print("Skewness:", df[quant].skew().round(3).to_dict())
    print("Kurtosis:", df[quant].kurt().round(3).to_dict())
    print("\nCorrelation matrix:\n", df[quant].corr().round(3))

    print("\n===== QUALITATIVE ANALYSIS (categorical data) =====")
    for c in qual:
        print(f"\n{c} - counts and percentages")
        print(pd.DataFrame({"Count": df[c].value_counts(),
                           "Percent": (df[c].value_counts(normalize=True) * 100).round(1)}))
    print("\nCross-tabulation Dept vs Satisfaction:\n", pd.crosstab(df["Dept"],
                           df["Satisfaction"]))
    print("\nAverage Salary by Dept:\n", df.groupby("Dept")["Salary"].mean().round(0))

    fig, ax = plt.subplots(2, 3, figsize=(15, 8))
    sns.histplot(df["Salary"], kde=True, ax=ax[0, 0]).set_title("Salary distribution")
    sns.boxplot(x="Dept", y="Salary", data=df, ax=ax[0, 1]).set_title("Salary by Dept")
    sns.heatmap(df[quant].corr(), annot=True, cmap="coolwarm", ax=ax[0, 2],
                2]).set_title("Correlation")
    sns.countplot(x="Dept", data=df, ax=ax[1, 0]).set_title("Dept counts")
    sns.countplot(x="Satisfaction", hue="Gender", data=df, order=["Low", "Medium", "High"],
                  ax=ax[1, 1]).set_title("Satisfaction by Gender")
    ax[1, 2].pie(df["Dept"].value_counts(), labels=df["Dept"].value_counts().index,
                 autopct="%1.1f%%")
    ax[1, 2].set_title("Dept share")
    plt.tight_layout()
    plt.show()
```

exp_16()

Output

```
===== QUANTITATIVE ANALYSIS (numerical data) =====
      Age      Salary  Experience
count  200.00    200.00    200.00
mean    39.20   62485.64     15.20
std     12.05   14862.07      8.64
min     18.00   22217.00      0.00
25%     29.00   53865.00      8.00
```

50%	39.50	62495.50	16.00
75%	50.00	71981.50	23.00
max	59.00	107412.00	29.00

Median : {'Age': 39.5, 'Salary': 62495.5, 'Experience': 16.0}
Mode of Age : 40
Skewness: {'Age': -0.045, 'Salary': -0.059, 'Experience': -0.159}
Kurtosis: {'Age': -1.192, 'Salary': -0.145, 'Experience': -1.218}

Correlation matrix:

	Age	Salary	Experience
Age	1.000	0.056	0.049
Salary	0.056	1.000	0.577
Experience	0.049	0.577	1.000

===== QUALITATIVE ANALYSIS (categorical data) =====

Dept - counts and percentages

	Count	Percent
Dept		
IT	76	38.0
Sales	47	23.5
HR	44	22.0
Finance	33	16.5

Gender - counts and percentages

	Count	Percent
Gender		
Male	102	51.0
Female	98	49.0

Satisfaction - counts and percentages

	Count	Percent
Satisfaction		
Medium	123	61.5
High	42	21.0
Low	35	17.5

Cross-tabulation Dept vs Satisfaction:

Satisfaction	High	Low	Medium
Dept			
Finance	5	9	19
HR	17	7	20
IT	15	7	54
Sales	5	12	30

Average Salary by Dept:

Dept	Average Salary
Finance	67041.0
HR	61362.0
IT	64131.0
Sales	57678.0

Name: Salary, dtype: float64

Graph

